VIVEKA-INSIGHT: A Cross-Lingual Concept-Graph Retrieval System and Citation-Grounded Question Answering over the Complete Works of Swami Vivekananda

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Abstract

Classical spiritual and philosophical corpora pose three compounding challenges for modern retrieval systems: they exist in multiple languages without parallel alignment, their vocabulary differs sharply from the vocabulary of contemporary readers, and generated answers over such culturally sensitive material must be verifiably grounded in the source text. We present VIVEKA-INSIGHT, an end-to-end open system that addresses all three for the works of Swami Vivekananda (1863–1902): the nine-volume English *Complete Works* and the ten-volume Bengali *Vani o Rachana*, two related but non-parallel corpora totalling about 15 million characters. The system combines (i) multi-granular dense retrieval at sentence, paragraph, and chapter level using a single multilingual embedding space; (ii) an LLM-extracted *concept graph* of 8,362 language-agnostic concepts with 87,499 paragraph–concept and 55,865 concept–concept edges, which serves as a shared abstraction layer bridging the two languages; and (iii) a question-answering pipeline that explicitly *bridges modern questions* (e.g., "how do I get rid of mobile phone addiction?") to era-appropriate concepts such as *attachment* and *self-control* before retrieval, then generates answers in which every claim carries a citation deep-linked to the exact source paragraph, with unsupported citation markers removed. The full index is built for under one GPU-day and serves interactive queries in 0.3 seconds and cited answers in 6–9 seconds on a single GPU. In a known-item cross-lingual evaluation over 194 automatically verified English–Bengali rendered lecture pairs, the production configuration retrieves the counterpart chapter at Recall@10 of 0.86 in both directions, against 0.03 for a BM25 baseline that cannot cross the language boundary; over a 30-question audit, the bridging stage fires on 93% of modern questions versus 13% of timeless ones, and no generated citation marker pointed outside the retrieved evidence. A human study of 200 randomly sampled paragraph–concept edges places concept-extraction precision at 0.60 under the strict consensus of two annotators (Cohen's $\kappa = 0.61$), and shows that the extractor's own confidence weight is calibrated: restricting the graph to weight ≥ 0.8 raises consensus precision to 0.71 while still covering 98% of the paragraphs that carry any concept. Precision is markedly lower on Bengali than on English material (0.54 vs 0.68), locating the concept layer's weakness in exactly the half of the corpus that cross-lingual bridging most depends on. We describe the architecture, report these studies alongside corpus and graph statistics, and present case studies of temporal concept bridging in both English and Bengali. The design generalizes to other multilingual classical corpora.

1 Introduction

Large digitized editions of classical religious and philosophical literature are now widely available, but access to them remains largely lexical: readers search for words, while what they hold are

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questions. The gap is widest for corpora like the works of Swami Vivekananda, a central figure of modern Vedanta whose collected English works span nine volumes and whose Bengali writings and recorded conversations fill ten more. The two collections overlap thematically but are *not* translations of one another; a reader fluent in only one language has no lexical route into roughly half of the material.

A second, subtler gap is *temporal.* Contemporary readers bring contemporary questions — about smartphone addiction, social-media envy, or career anxiety — to an author who died in 1902. The vocabulary of such questions simply does not occur in the corpus, so even strong dense retrievers underperform; yet the corpus deals extensively with the underlying phenomena under names such as *attachment*, *control of the mind*, and *habit.* Bridging this gap is not a retrieval nicety but the central requirement for making a nineteenth-century corpus useful to twenty-first-century readers.

Finally, any system that *generates* answers about revered texts must confront hallucination [13]. Attributing invented statements to a religious figure is a materially worse failure mode than a wrong answer in open-domain QA, so verifiable grounding — every claim traceable to a specific passage a reader can open — is a hard constraint rather than a desirable extra.

VIVEKA-INSIGHT is a complete, deployed system built around these three requirements. Its contributions are:

1. **A cross-lingual concept graph as retrieval infrastructure.** An instruction-tuned LLM reads every paragraph of both corpora and emits concepts with *canonical English, lowercase, hyphenated labels.* Because the label space is shared, a Bengali paragraph about *vairagya* and an English lecture on renunciation attach to the same node; string equality on labels gives cross-lingual linking without any parallel data (Section 4).

2. **Multi-granular fused retrieval.** Queries are matched against sentence-, paragraph-, and chapter-level indexes in both languages *and* against the concept graph; candidates are fused with weighted reciprocal rank fusion [8] and reranked by a multilingual cross-encoder (Section 5).

3. **Temporal concept bridging for QA.** A dedicated analysis step maps modern questions onto the concept vocabulary that actually exists in the graph before retrieval, and the mapping is surfaced to the user as an explicit, inspectable "bridge" (Section 6).

4. **Citation-grounded generation.** Answers are composed only from retrieved passages; each claim carries a bracketed citation that is rewritten into a deep link to the exact *paragraph* of the published source, and citation markers that do not correspond to a retrieved passage are deleted (Section 6).

5. **An economical, reproducible build.** The full pipeline runs on a single GPU in under a day, incremental rebuilds after source edits take 10–15 minutes via a warm-start cache, and the system serves interactive latencies on modest hardware (Section 7).

2 Related Work

Dense retrieval and RAG. Dense passage retrieval [2] and retrieval-augmented generation [1, 3] are the standard architecture for grounded QA. We adopt the general recipe but depart in two ways: the retrieval layer is multi-granular and graph-augmented, and generation is constrained to a citation discipline enforced by post-processing rather than trust in the model.

Graph-augmented retrieval. GraphRAG [4] builds an entity graph over a corpus with an LLM and uses community summaries for query-focused summarization. VIVEKA-INSIGHT shares the LLM-builds-a-graph idea but uses the graph differently: nodes are *concepts* with

Table 1: Corpus statistics after parsing.

	English	Bengali	Total
Volumes (published)	9	10	19
Chapters	1,452	539	1,991
Paragraphs	18,239	14,455	32,694
Sentences	88,150	80,692	168,842
Characters	8.1M	6.9M	15.0M

language-agnostic canonical labels, and the graph's primary role is cross-lingual bridging and query-time expansion into two non-parallel corpora, rather than corpus summarization.

Multilingual embeddings. We build on BGE-M3 [5], whose single embedding space covers both English and Bengali and supports inputs long enough for chapter-level representation, and on its companion multilingual cross-encoder reranker. Sentence-level transformer embeddings follow the general approach of [6].

QA over religious corpora. Verse-based test collections and QA systems exist for the Qur'an [17] and other scriptures; these typically operate in a single language over a closed canon. To our knowledge, VIVEKA-INSIGHT is the first system to combine non-parallel bilingual retrieval, an LLM-built concept layer, and explicit modern-question bridging for a classical corpus.

3 Corpus and Preprocessing

The English source is the nine-volume *Complete Works of Swami Vivekananda* (lectures, letters, recorded conversations, and writings; ∼8.1 million characters). The Bengali source is the ten-volume *Vani o Rachana* (∼6.9 million characters), which contains original Bengali writings and conversations as well as independent Bengali renderings of some English lectures; no paragraph-level alignment between the two exists, and we do not attempt to create one.

Both corpora are parsed from their published HTML editions with structure-aware parsers that recover the volume → (section) → chapter → paragraph hierarchy and preserve both the chapter and the per-paragraph HTML anchor identifiers, which later enable paragraph-precise deep-linking of citations back into the publicly hosted editions. Sentences are split with language-appropriate rules (NLTK [12] for English; a rule-based splitter honouring the *danda* and Bengali punctuation for Bengali). Bengali text is NFC-normalized throughout; a single normalization function is the sole source of truth for text keys across the pipeline, which the incremental-rebuild machinery (Section 4.2) depends on. Table 1 summarizes the corpus.

4 The Concept Graph

4.1 LLM concept extraction

An instruction-tuned LLM (Qwen2.5-14B-Instruct [9], served with vLLM [10]) reads every paragraph in both languages and emits structured JSON containing (i) a one-line English summary, (ii) a set of *concepts*, and (iii) a set of named *entities* typed as person, place, text, or deity. The extraction prompt imposes the single most important invariant of the system: **concept labels are canonical English, lowercase, and hyphenated** regardless of the paragraph language. The canonical label thus acts as a cross-lingual key: the Bengali surface form and the English surface form of a concept are stored as *aliases* (60,838 aliases; 30,051 English, 30,787 Bengali), while identity lives in the label itself. Each paragraph–concept edge carries a relation (*discusses*, *exemplifies*, or *contrasts*) and a weight.

Near-duplicate concepts produced by paraphrase (e.g., spelling variants) are merged by

Table 2: Concept-graph statistics.

Concepts (canonical labels)	8,362
Concept aliases (EN / BN)	30,053 / 30,797
Entities (person / other / place / text / deity)	3,688 / 2,372 / 1,275 / 846 / 492
Paragraph→concept edges	87,518
discusses / exemplifies / contrasts	75,403 / 8,735 / 3,380
Concept↔concept edges	55,872
similar / co-occurs	29,482 / 26,390
Paragraphs with ≥1 concept	23,363 (71.5%)
retained at extractor weight ≥ 0.8	22,914 (98.1%)

Table 3: Most frequent concepts by paragraph mentions.

Concept	Mentions	Concept	Mentions
self-realization	4,521	renunciation	2,553
knowledge-realization	3,858	liberation	2,528
devotion	3,179	patience	1,787
maya	2,684	duty	1,437
non-attachment	2,669	karma	1,334

embedding the labels and clustering above a conservative cosine threshold (0.92). Two kinds of concept–concept edges are then built: *similar* edges (label-embedding cosine $\geq$ 0.78, top-12 neighbours per node) and *co-occurs* edges (concepts extracted from the same paragraph at least twice corpus-wide). Table 2 summarizes the resulting graph; Table 3 lists the most frequent concepts, which align well with the corpus's known thematic centres.

The 71.5% paragraph coverage is deliberate rather than a deficiency: short fragments (single-word replies in recorded conversations, headings, brief exclamations) legitimately carry no concept content, and forcing extraction on them degrades graph precision.

Coverage, however, is not correctness. A two-annotator study of the edges themselves (Section 8.3) puts their strict-consensus precision at 0.60, rising to 0.71 on the high-confidence subgraph and falling to 0.54 on Bengali material; the graph should be read as a broad, imperfect expansion layer rather than a curated index, and everything we claim for it below is calibrated to that.

4.2 Economical construction and incremental rebuilds

Concept extraction is the only expensive stage ($\sim$3.5 hours for the full corpus on a single 80 GB-class GPU); everything else (parsing, embedding, graph assembly) takes minutes. To make the corpus *editable* — OCR fixes and structural cleanups are routine in digitized classical texts — the pipeline snapshots all extraction outputs keyed by (NFC-normalized paragraph text, language, extractor version) before any re-parse, and restores them onto the fresh paragraph rows afterwards. Only genuinely new or changed paragraphs reach the LLM. An incremental rebuild after a content edit completes in 10–15 minutes, which in practice is the difference between a maintained resource and a frozen one.

5 Multi-Granular Fused Retrieval

All text units — 168k sentences, 33k paragraphs, 2k chapters, and the 8.4k concept labels — are embedded into the same 1024-dimensional space with BGE-M3 [5] and stored in exact

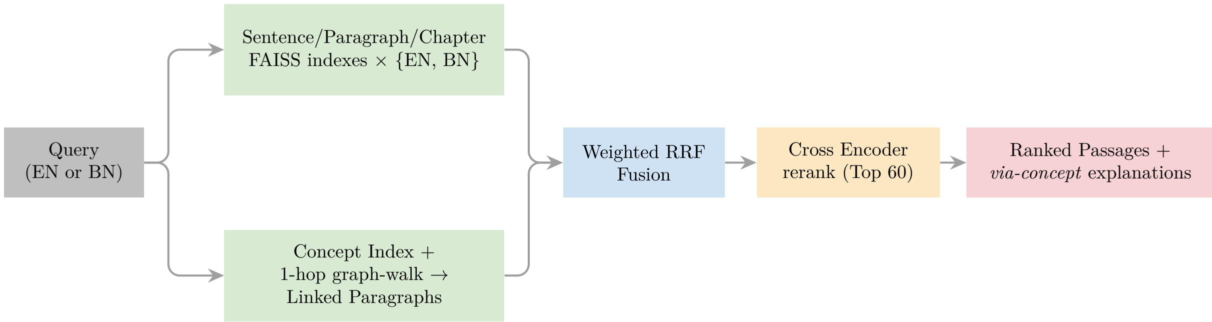

Figure 1: Retrieval architecture. Path A (top) matches the query against three granularities per language; Path B (bottom) routes it through the concept graph. Both feed a single fusion and reranking stage.

inner-product FAISS indexes [7] (about 1.1 GB in total including metadata, small enough that approximate indexing is unnecessary). Retrieval runs two paths per language (Figure 1):

Path A (direct). The query is matched against the sentence (top-30), paragraph (top-20), and chapter (top-10) indexes. Sentence and chapter hits are mapped to their enclosing/leading paragraphs, since the paragraph is the unit surfaced to users; the granularity that found each hit is retained and displayed.

Path B (concept-mediated). The query is matched against the concept-label index (top-12 anchors), the anchors are expanded one hop along *similar* and *co-occurs* edges with weight decay, and each activated concept contributes its top linked paragraphs. Because concept nodes are language-agnostic, an English query activates Bengali paragraphs (and vice versa) with no translation step. Path B also yields an explanation — the concepts through which each paragraph was reached — which the interface displays as "via concepts" badges.

Candidates from the four rank lists (three granularities + concept path) are fused with weighted reciprocal rank fusion [8], with the concept path weighted slightly above the direct paths (1.2 vs. 1.0) and chapters below (0.7). The top 60 fused candidates are reranked by the BGE-reranker-v2-m3 cross-encoder, and the top k per language are returned with full provenance (volume, section, chapter, paragraph index, and a deep link to the chapter anchor in the hosted edition).

6 "Ask Vivekananda": Bridged, Citation-Grounded QA

The QA layer turns retrieval into direct answers under two hard constraints: modern questions must reach era-appropriate material, and every generated claim must be verifiable. The pipeline has three stages.

Stage 1: analyze and bridge. The question's 15 nearest concept labels are fetched from the concept index. A single LLM call receives the question *and this list* and returns (i) 2–4 reformulated "timeless" search queries free of modern-only vocabulary, (ii) up to 6 concept labels *copied from the offered list*, and (iii) a one-line *bridge note* explaining the modern-to-timeless mapping. Restricting concept choice to labels that verifiably exist in the graph (validated again against the database after generation) prevents the bridge itself from hallucinating; if JSON parsing fails, the system degrades to retrieval with the raw question. The question's language is detected deterministically from its script, never delegated to the LLM.

Stage 2: retrieve. The original question and each reformulation are run through the full retrieval stack of Section 5; results are merged, deduplicated per (language, paragraph), and the pooled candidates are reranked once against the *original* question. The top 10 passages become the numbered sources.

Stage 3: generate and ground. A locally hosted Qwen2.5-7B-Instruct [9] (served via

Table 4: Serving-time characteristics (single NVIDIA H200; the system also runs CPU-only with seconds-level search latency). Model download sizes excluded.

Search, with cross-encoder rerank (mean of 5 queries)	0.28 s
Search, fusion only	0.08 s
QA end-to-end (bridge + retrieve + generate)	6–9 s
Index footprint (7 FAISS indexes + SQLite metadata/graph)	1.1 GB
Full pipeline build (one GPU)	< 4 h
Incremental rebuild after a source edit	10–15 min

Transformers [11]; any OpenAI-compatible endpoint can be substituted by configuration) receives the numbered sources — location-labelled, language-tagged, and trimmed to a window centred on each passage's best-matching sentence — and composes a scholarly third-person answer in the language of the question. The system prompt forbids claims not supported by the sources and requires a bracketed citation after every substantive statement. Post-processing then rewrites each citation marker [n] into a hyperlink into the published online edition and *deletes* any marker whose index does not correspond to a retrieved source, so a fabricated citation cannot survive to the reader. The published editions carry a stable HTML anchor on every paragraph, so each citation deep-links to the *exact cited paragraph* (falling back to the chapter anchor where a paragraph anchor is absent) — the reader lands on the sentence in question, not merely the top of a chapter. The interface additionally shows the bridge note, the reformulated queries, the chosen concepts, and, for every source, the matched sentence and the full passage — the entire evidential chain is one click deep.

7 System Characteristics

The stack is deliberately conservative: exact FAISS search, SQLite for all metadata and graph storage, and a Streamlit interface. There are no external service dependencies at query time; the system runs entirely on-premises, which matters both for cost and for institutions that cannot ship religious-community data to third-party APIs. Table 4 reports measured characteristics. A test suite of 74 CPU-only tests covers the parsers, schema, extraction-output parsing, fusion, bridging fallbacks, prompt budgeting, citation linkification, and the concept-graph view.

8 Evaluation

We report three studies: two automatic ones that require no manual annotation, and a human study of the concept layer. All are reproducible with evaluation scripts shipped in the repository.[1]

8.1 Known-item cross-lingual retrieval

The *Vani o Rachana* contains independent Bengali renderings of many English lectures, which yields chapter-level cross-lingual relevance judgments at near-zero annotation cost. We identified candidate pairs as mutual-best matches between the chapter-level embedding indexes (cosine ≥ 0.70; 285 candidates), then verified each candidate with the local LLM shown both chapters' titles and opening paragraphs, retaining $N = 194$ verified rendered pairs. For each pair, the source chapter's title and opening paragraph (truncated to 600 characters) is the query, retrieval is restricted to the *other* language, and the paired chapter is the sole gold item; a retrieved

[1] `scripts/eval_crosslingual.py`, `scripts/eval_qa_integrity.py`, and `scripts/eval_concept_precision.py`; the raw outputs quoted here, including the per-annotator judgment files, are included under `docs/paper/eval/`.

Table 5: Known-item cross-lingual retrieval over $N = 194$ LLM-verified rendered lecture pairs (EN→BN / BN→EN). The upper block is external baselines; the lower block ablates our own stack. The last row is the production default.

Configuration	Recall@10	MRR
Baselines		
BM25 (lexical)	0.03 / 0.02	0.01 / 0.02
Dense paragraph retrieval	0.88 / 0.81	0.71 / 0.62
Entity graph (GraphRAG-style)	0.20 / 0.08	0.12 / 0.03
This system		
Path A (direct dense, multi-granular)	0.89 / 0.82	0.74 / 0.65
Path B (concept-mediated)	0.14 / 0.12	0.07 / 0.04
Fused (weighted RRF)	0.89 / 0.82	0.68 / 0.61
Fused + cross-encoder rerank	0.86 / 0.86	0.67 / 0.69

paragraph counts if it belongs to the gold chapter. We report Recall@10 and MRR over the top 10 returned paragraphs in both directions (Table 5).

Two caveats. Pair discovery uses the same embedder as Path A (albeit on whole-chapter vectors, while queries are titles plus opening paragraphs), so absolute dense numbers may be slightly optimistic; comparisons *across* configurations are unaffected, and the BM25 baseline is independent of that bias. Second, the verified pair set is cached, and a full re-parse reassigns chapter row identifiers; the evaluation script detects and repairs stale identifiers by chapter title so that the cached LLM verdicts remain usable across rebuilds.

Lexical retrieval cannot do this task at all. BM25 reaches Recall@10 of 0.03 and 0.02. That is not a weak baseline but a floor at chance: English and Bengali share almost no surface vocabulary, so term matching has nothing to match on. It is worth stating plainly, because it establishes that every other number in the table is bought by *semantic* representation rather than by any lexical signal,.

Most of the known-item performance comes from the multilingual embedder, not from our additions. Plain paragraph-level dense retrieval — one index, no fusion, no reranking, no graph — already reaches 0.88 / 0.81. Adding sentence- and chapter-level indexes (Path A) is worth about a point of recall and 0.03 MRR. We report this because it bounds the claim honestly: the contribution of this work is not that it retrieves known items better than a competent dense baseline, but that it adds cross-lingual *explanation*, query bridging and citation grounding on top of one, at no cost to that baseline's accuracy.

Neither graph path is a known-item retriever, and the entity graph is not better than the concept graph at being one. Routing retrieval through the entity layer — the closest available stand-in for a GraphRAG-style system, built by the same extraction pass over the same corpus — gives 0.20 / 0.08, against 0.14 / 0.12 for the concept path. The two are comparable and both are far below the dense paths, which is what should be expected: graph expansion widens a query into thematically related material, and known-item metrics penalise exactly that. The distinction that matters between the two layers is not visible in this table. It is that entity nodes are proper nouns, so an entity graph links passages that mention the same person or place, whereas concept nodes carry the argument, and their canonical English labels give a cross-lingual key that entity strings in two scripts do not. What Path B buys is reach and the "via concepts" explanations, not ranking.

Fusion preserves recall and the cross-encoder balances the directions. Fusion keeps Path A's recall while admitting the concept candidates, and the cross-encoder markedly improves the harder BN→EN direction (MRR 0.61→0.69, Recall@10 0.82→0.86) at a small cost in EN→BN, yielding the most balanced configuration — the production default.

Table 6: Citation-integrity and bridging statistics over 30 questions (20 EN / 10 BN; 15 modern / 15 timeless).

Questions answered (non-empty retrieval)	30 / 30
Stage-1 JSON fallback rate	0%
Bridge note produced: modern questions	14 / 15 (93%)
Bridge note produced: timeless questions	2 / 15 (13%)
Citation markers per answer sentence	0.47
Citation markers emitted (total)	136
Fabricated markers removed by validator	0 / 136 (0%)
Cross-lingual share of top-10 evidence (bridged)	32%
Cross-lingual share of top-10 evidence (raw question)	40%

8.2 Citation integrity and bridge reliability

We ran the full QA pipeline over a fixed set of 30 questions — 20 English and 10 Bengali; 15 phrased in deliberately modern vocabulary (smartphone addiction, social-media envy, job-loss anxiety, burnout) and 15 in timeless vocabulary (fear, concentration, duty, anger) — and logged the pipeline's internal decisions (Table 6).

The bridging stage discriminates almost perfectly between question types: it produced a bridge note for 93% of modern questions and only 13% of timeless ones, with a 0% structured-output failure rate. Generated answers cite densely (one marker per two sentences on average), and in this run the 7B model emitted *no* citation marker pointing outside the retrieved source list — the validator, which deletes such markers, acted as an idle guardrail rather than an active repair mechanism.

The bridge ablation produced a finding we did not anticipate: the cross-lingual share of the evidence pool is essentially unchanged by bridging (32% bridged vs. 40% raw; 47% vs. 45% on the modern subset). Cross-linguality is supplied by the shared embedding space and the concept graph themselves, not by the reformulation step. The bridge's measured contributions are therefore selective activation (firing when and only when the vocabulary gap exists), the era-appropriate reformulations, and the user-facing explanation — not an increase in cross-lingual recall, which the underlying retrieval already provides.

8.3 Concept-extraction precision: a human study

The concept layer is the system's central claim, and it is produced entirely by an LLM. We therefore ran a human study of its precision.

Protocol. From all 87,518 paragraph–concept edges we drew a uniform random sample of $N = 200$ (seed 13). Each item presents one paragraph, one attached concept label, and the extracted relation. Annotators judged each item as *correct* (the concept is a correct reading of the paragraph *at the stated relation*) or *incorrect* (wrong, merely tangential, or right concept but wrong relation). Judgments were collected through a small purpose-built web interface shipped with the repository, which saves after every click and lets an annotator resume; no annotator saw another's judgments or the extractor's confidence weight.

Annotators. Three annotators, all familiar with the corpus, each judged all 200 items independently. Annotators A and B are fluent in both English and Bengali; annotator C is proficient principally in Bengali. Since 85 of the 200 sampled items are English, we use C's judgments only for the Bengali subsample, where that proficiency applies, and base the whole-sample figures on A and B. All three judgment files ship with the repository.[2]

[2] `docs/paper/eval/annotations/`; the whole-sample figures below are reproduced with `eval_concept_precision.py compute -users annotator_a annotator_b`, and any other pairing can be recomputed from the same files.

Table 7: Concept-extraction precision over $N = 200$ randomly sampled paragraph–concept edges, annotators A and B. Intervals are Wilson 95% (bootstrap for κ).

Measure	Value	95% CI
Precision, annotator A (non-author)	0.665	[0.597, 0.727]
Precision, annotator B (author)	0.705	[0.638, 0.764]
Precision, consensus (both correct)	**0.600**	[0.531, 0.665]
Precision, union (either correct)	0.770	[0.707, 0.823]
Raw agreement	0.830	—
Cohen's κ [14]	0.607	[0.482, 0.720]
Gwet's AC_1 [16]	0.701	—

Table 8: Precision by extractor confidence weight. The high-confidence subgraph is 35.9% of all edges but still covers 98.1% of the paragraphs that carry any concept.

	Annot. A	Annot. B	Consensus
weight ≥ 0.8 $(n = 69)$	0.783	0.812	**0.710**
weight < 0.8 $(n = 131)$	0.603	0.649	0.542

That scoping is a competence requirement fixed by the task, and the measured reliability tracks it closely: against A, annotator C reaches $\kappa = 0.49$ on the Bengali items but only $\kappa = 0.14$ — barely above chance — on the English ones. Restricting C to Bengali is therefore not a way of setting aside an inconvenient judge but of using each annotator where they are reliable; we report the Bengali three-annotator panel in full below, and it does not flatter the system.

Disclosure. Annotator B is an author of this paper; annotators A and C are not. B is marginally the more lenient of A and B (0.705 vs 0.665), so we treat the strict consensus figure — not B's — as the headline number.

Precision is moderate; agreement is substantial. Consensus precision is 0.60 — three in five sampled edges are endorsed by both judges — and Cohen's κ of 0.61 falls in the "substantial" band on the conventional scale [15]. The residual disagreement is not systematic: 21 items go one way and 13 the other, which a McNemar exact test does not distinguish from chance ($p = 0.23$). The two are applying the same rubric with ordinary noise, which is what an inter-annotator agreement study is supposed to establish before its precision figure is worth quoting.

The extractor's confidence weight is calibrated. Each paragraph–concept edge carries a weight the extraction LLM assigns itself. It predicts human judgment (Spearman $\rho = 0.19$ against consensus), and thresholding on it works (Table 8): consensus precision rises from 0.54 below 0.8 to 0.71 at 0.8 and above (Fisher exact $p = 0.023$).

The pruning is nearly free in coverage terms. Applied to the full graph, weight ≥ 0.8 retains 31,445 of 87,518 paragraph–concept edges (35.9%) and 3,709 of 8,362 concepts (44.4%) — but 22,914 of the 23,363 paragraphs that carry any concept (**98.1%**) keep at least one. The low-confidence mass is redundant attachment to already-covered paragraphs, not the sole handle on distinct material. A deployment that needs the graph to be *right* rather than *broad* should therefore threshold at 0.8; we report the unpruned graph throughout this paper because Path B is used for recall-oriented expansion, where the cost of a wrong concept is a wasted glance rather than a false statement.

Extraction is weaker in Bengali. Consensus precision over A and B is 0.68 on English items and 0.54 on Bengali (Fisher exact $p = 0.043$). The Bengali subsample ($n = 115$) is the one part of the study carrying a third independent judgment, and the fuller panel confirms the gap rather than closing it: across A, B and C the three-rater Fleiss κ is 0.49 and majority-of-three precision is 0.59, still well below the 0.68 measured on English. Adding a Bengali-proficient annotator thus corroborates the finding with the language's strongest available judgement.

This is the study's one clearly actionable quality finding beyond the threshold: the extraction LLM is a predominantly English-trained model reading Bengali prose, and the concept layer inherits that asymmetry. Since cross-lingual bridging is the system's central claim, the weaker half of the graph is precisely the half doing the work of making Bengali material reachable from English queries — a Bengali-strong extractor, or a second extraction pass over Bengali paragraphs alone, is the most valuable single upgrade available to the pipeline.

Relations. The dominant *discusses* (87.5% of the sample) reaches consensus precision 0.62, while the two rare relations are worse (*exemplifies* 0.47, $n = 17$; *contrasts* 0.50, $n = 8$) — on samples too small to support a strong claim, but consistent with the intuition that a directional judgment is harder for the extractor than a topical one.

On the rubric. The binary judgment conflates two readings of "the concept describes the paragraph": whether the paragraph *is about* the concept, and whether the concept is merely *present* in it. For a Sūtra gloss such as "The success of Yogis differs according as the means they adopt are mild, medium, or intense" tagged `success`, or a Bengali travel note announcing a forthcoming Vedanta lecture tagged `vedanta`, the first reading rejects and the second accepts. Both annotators converged on the more permissive presence reading, so the 0.60 we report should be understood as a precision under that reading; a study that fixed the stricter aboutness criterion for every annotator would return a lower number. A revised study should use a three-point scale (*central* / *mentioned* / *unsupported*) rather than force this distinction into a binary, and should fix the reading in the guidelines instead of leaving it to be resolved per annotator.

What the errors look like. The characteristic failure is not a bizarre concept but a plausible one attached to a paragraph that does not support it. Narrative, epistolary, and connective passages attract the corpus's thematic vocabulary regardless of content: a Bengali letter remarking that a book "has been well received here, but the publishers do not seem to be pushing sales" (our translation) is tagged `knowledge-realization`; a subscription appeal for a memorial is tagged `liberation`. The extractor appears to condition on the corpus's overall register as much as on the paragraph in front of it. This is precisely the failure mode that the confidence threshold suppresses.

Implication for the system's claims. The concept layer should be understood as a *recall-oriented, imperfect expansion and explanation layer*, not a curated ontology. This is consistent with Table 5, where Path B is a poor known-item retriever on its own; its contribution is cross-lingual reach and the "via concepts" explanations, both of which degrade gracefully under error because every retrieved passage is displayed to the reader with its source. A wrong concept costs an irrelevant result, not a false assertion about Vivekananda — the citation discipline of Section 6, not the concept graph, is what bounds that risk.

9 Case Studies

We illustrate the bridging behaviour with unedited system outputs.

Modern question, English. For *"How do I get rid of mobile phone addiction?"* the analysis stage produced the bridge note *"Mobile-phone addiction is a modern form of attachment of the senses and a habit that enslaves the mind"*, selected the corpus concepts `excessive-attachment` and `self-control`, and reformulated the question into queries such as *"How can I overcome excessive attachment to sense objects?"* and *"What is the remedy for habit that enslaves the mind?"*. Retrieval returned ten passages spanning *Inspired Talks*, the Raja-Yoga lectures, and — notably — Bengali chapters on liberation (*mukti*) and sense-withdrawal (*pratyahara*), which no lexical and no unbridged dense query for "mobile phone" reaches. The generated answer opens by restating the bridge, then prescribes non-attachment and graded practice of sense-withdrawal, with every paragraph citing its sources.

Modern question, second example. For *"How should one deal with social media envy?"*

the bridge note read *"Social media envy is a modern form of coveting the social status and approval of others. . . "* — a mapping to covetousness and approval-seeking that the authors did not anticipate but that the concept inventory supported.

Bengali question. A Bengali question on freedom from mobile addiction (*"mobile asakti theke muktir upay ki?"*) was detected as Bengali, bridged through the same English-labelled concepts, answered *in Bengali*, and grounded predominantly in Bengali sources while still drawing on English lectures — the concept layer making the cross-lingual evidence pool available in both directions.

10 Limitations and Ethical Considerations

An imperfect concept layer. The human study of Section 8.3 reports strict-consensus precision of 0.60 over paragraph–concept edges, and 0.54 on Bengali material. The concept graph is a useful expansion and explanation layer, but it is not a scholarly index and should not be cited as one. Users who need higher precision should threshold at extractor weight ≥ 0.8, which lifts consensus precision to 0.71 at almost no cost in paragraph coverage.

Limits of that study. The whole-sample figure rests on two annotators, one of them an author of this paper and marginally the more lenient of the two, judging 200 items — enough to place the precision within roughly ± 0.07, but not to support fine comparisons. A third annotator, proficient principally in Bengali, contributes only to the Bengali subsample; the English half therefore carries no judgement from outside the author's immediate circle. The binary rubric conflated "the paragraph is about this concept" with "this concept appears in the paragraph" and left the choice to each annotator; the two whole-sample annotators settled on the more permissive reading, so the headline figure is the optimistic end of the plausible range — the Bengali-proficient annotator, who judged more strictly, would place it lower. A three-point scale, guidelines that fix the reading, and a larger and wholly independent annotator pool are all needed before the precision figure can be treated as settled rather than indicative.

Answer faithfulness is still unevaluated. Citation integrity is measured structurally (marker validity), not semantically: we verify that every citation points at a retrieved passage, not that the passage supports the claim attached to it. Known-item judgments likewise come from rendered lecture pairs rather than topical relevance assessment. A scholar-annotated faithfulness study of generated answers remains the most important piece of future work.

Generation quality floor. The default 7B answer model occasionally paraphrases loosely when quoting, and may present material from a Bengali source in English without consistently marking it as translated, despite prompt instructions. The citation mechanics bound the damage — the linked passage is always one click away — but do not eliminate it. Configurable substitution of a stronger model raises the floor.

Extraction bias. The concept inventory reflects the judgement of the extraction LLM; systematic blind spots in that model's reading of Vedantic material would propagate to the graph. The alias tables and per-paragraph provenance make audits feasible.

Cultural sensitivity. The corpus is scripture-adjacent for a living community. We consider verifiable grounding, the visible bridge explanation, and on-premises deployment to be ethical requirements for this material, not merely engineering choices, and we would caution against deploying fluent-generation systems over such corpora without citation enforcement of comparable strictness.

11 Conclusion

Viveka-Insight demonstrates that a modest, fully local pipeline — one extraction pass by a mid-sized LLM, a shared multilingual embedding space, and a canonical-label concept graph — suffices to make a bilingual, non-parallel classical corpus searchable by meaning, answerable in either language, and honest about its evidence. The temporal bridge turns the system's most obvious weakness (a nineteenth-century vocabulary) into an explicit, inspectable mapping rather than a silent retrieval failure. The architecture assumes nothing Vivekananda-specific: any corpus with stable canonical structure and one or more languages covered by the embedding model — the Ramakrishna *Kathamrita* literature, Gandhi's collected works in English, Gujarati and Hindi, or patristic corpora in Greek and translation — could be indexed with the same pipeline. The human study also marks the clearest directions for the next iteration. At strict-consensus precision of 0.60 the concept layer is broad but imperfect, and two of its properties point at cheap gains: the extractor's own confidence weight separates good edges from bad ones well enough that a calibrated threshold buys accuracy almost for free, and the precision gap between English (0.68) and Bengali (0.54) says the largest single quality deficit is not the pipeline but the extraction model's weaker reading of Bengali — the very half of the corpus that cross-lingual bridging exists to open up. A Bengali-capable extractor and community-driven correction through the audit trail the system already exposes are therefore the first steps; graph-embedding refinement of the concept layer and a scholar-annotated faithfulness study of generated answers are the further ones.

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References

[1] P. Lewis, E. Perez, A. Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.

[2] V. Karpukhin, B. Oğuz, S. Min, et al. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.

[3] Y. Gao, Y. Xiong, X. Gao, et al. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.

[4] D. Edge, H. Trinh, N. Cheng, et al. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.

[5] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu. BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*, 2024.

[6] N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, 2019.

[7] J. Johnson, M. Douze, and H. Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

[8] G. V. Cormack, C. L. A. Clarke, and S. Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of SIGIR*, 2009.

[9] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[10] W. Kwon, Z. Li, S. Zhuang, et al. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023.

[11] T. Wolf, L. Debut, V. Sanh, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, 2020.

[12] S. Bird and E. Loper. NLTK: The natural language toolkit. In *Proceedings of the ACL Interactive Poster and Demonstration Sessions*, 2004.

[13] Z. Ji, N. Lee, R. Frieske, et al. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.

[14] J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

[15] J. R. Landis and G. G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977.

[16] K. L. Gwet. Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1):29–48, 2008.

[17] R. Malhas and T. Elsayed. AyaTEC: Building a reusable verse-based test collection for Arabic question answering on the Holy Qur'an. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 19(6):1–21, 2020.